MATH 231 – Linear Algebra I

The Matrix Unit, Part 9 ▪ Least Squares

Kiely Hall 326 ▪ Instructor: Kevin Bedoya

About this document. This is Part 9 of the matrix unit, Sections 46–49. Section numbers run continuously through the unit (Part 2's opening table lists them all). References into other units always name them.

Assumed. From the vector unit: projection (§12.4–§12.6) and the Pythagorean identity (§23). From this unit: the column space (§10), projection matrices and orthogonal projections (§26–§27), the null space and $\operatorname{rank}(\mathbf{A}^\mathsf{T}\mathbf{A}) = \operatorname{rank}(\mathbf{A})$ (§31–§33), and elimination (Part 8). From the vector calculus unit: the derivative (§1) and partial derivatives and the gradient (§6–§7).

What you should be able to do afterwards. Find the multiple of one vector closest to another, by calculus and by geometry; explain why the best approximate solution of $\mathbf{Ax} = \mathbf{b}$ makes the residual orthogonal to $\operatorname{col}[\mathbf{A}]$; derive the normal equations $\mathbf{A}^\mathsf{T}\mathbf{A}\hat{\mathbf{x}} = \mathbf{A}^\mathsf{T}\mathbf{b}$ both geometrically and by setting a gradient to zero; say when the least-squares solution is unique; and fit a line to data.

Part 8 ended with three data points that no line passes through, and an inconsistent system $\mathbf{Ax} = \mathbf{b}$ to say so. Every measured dataset does this: with more measurements than unknowns, and a little noise in each, the equations cannot all hold. The system has no solution, and yet some candidate answers are clearly better than others. *Least squares* chooses the one that makes the equations fail by as little as possible, in the sense of the Euclidean length.

46. The closest point on a line

46.1. The problem

Start with the smallest version, one unknown. Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ be vectors with $\mathbf{u} \neq \mathbf{0}$, and let x be a scalar. The vectors $x\mathbf{u}$ make up the line through $\mathbf{u}$. Which of them is closest to $\mathbf{v}$? We look for

$$\hat{x} = \operatorname*{arg\,min}_{x \in \mathbb{R}} \|\mathbf{v} - x\mathbf{u}\|_2^2,$$

the value of x at which $\|\mathbf{v} - x\mathbf{u}\|_2^2$ is smallest. Squaring the distance does not change where the minimum is, since a larger distance always has a larger square, and it removes the square root.

46.2. It is a parabola

Expand the squared length with $\|\mathbf{w}\|^2 = \mathbf{w}^\mathsf{T}\mathbf{w}$ (§7.3) and the distributive laws, using $\mathbf{v}^\mathsf{T}\mathbf{u} = \mathbf{u}^\mathsf{T}\mathbf{v}$:

$$f(x) = \|\mathbf{v} - x\mathbf{u}\|^2 = (\mathbf{v} - x\mathbf{u})^\mathsf{T}(\mathbf{v} - x\mathbf{u}) = \mathbf{v}^\mathsf{T}\mathbf{v} - 2x\,\mathbf{u}^\mathsf{T}\mathbf{v} + x^2\,\mathbf{u}^\mathsf{T}\mathbf{u}.$$

The numbers $\mathbf{v}^\mathsf{T}\mathbf{v}$, $\mathbf{u}^\mathsf{T}\mathbf{v}$, $\mathbf{u}^\mathsf{T}\mathbf{u}$ are fixed, so f is an ordinary quadratic in the single variable x. Its leading coefficient $\mathbf{u}^\mathsf{T}\mathbf{u} = \|\mathbf{u}\|^2$ is positive, so it is a parabola opening upward, with one lowest point.

46.3. The minimum

From the vector calculus unit. At the lowest point of a differentiable function of one variable, the derivative is zero (§1).

$$f'(x) = -2\,\mathbf{u}^{\mathsf{T}}\mathbf{v} + 2x\,\mathbf{u}^{\mathsf{T}}\mathbf{u} = 0 \qquad \Longrightarrow \qquad \boxed{\hat{x} = \frac{\mathbf{u}^{\mathsf{T}}\mathbf{v}}{\mathbf{u}^{\mathsf{T}}\mathbf{u}}.}$$

And $f''(x) = 2\,\mathbf{u}^{\mathsf{T}}\mathbf{u} > 0$, confirming a minimum.

46.4. It is the projection

That fraction is familiar. The closest point is

$$\hat{x}\,\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{u}\cdot\mathbf{u}}\,\mathbf{u} = \mathrm{proj}_{\mathbf{u}}(\mathbf{v}) = \mathbf{P}_{\mathbf{u}}\mathbf{v},$$

the projection of the vector unit's §12.4, which is the projection matrix of §26 applied to $\mathbf{v}$. The calculus has rediscovered the projection: *the point of a line closest to* $\mathbf{v}$ *is the foot of the perpendicular from* $\mathbf{v}$. The leftover, the **residual** $\mathbf{r} = \mathbf{v} - \hat{x}\mathbf{u}$, is the orthogonal piece of the vector unit's §12.6:

$$\mathbf{u}^{\mathsf{T}}\mathbf{r} = \mathbf{u}^{\mathsf{T}}\mathbf{v} - \hat{x}\,\mathbf{u}^{\mathsf{T}}\mathbf{u} = \mathbf{u}^{\mathsf{T}}\mathbf{v} - \mathbf{u}^{\mathsf{T}}\mathbf{v} = 0.$$

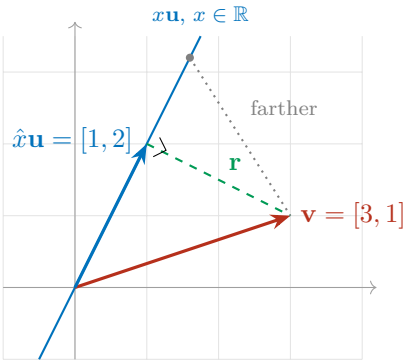
46.5. Why the perpendicular is closest, without calculus

Take any other x and write $\mathbf{v} - x\mathbf{u} = \mathbf{r} + (\hat{x} - x)\mathbf{u}$. The two pieces are orthogonal ($\mathbf{r} \perp \mathbf{u}$), so the Pythagorean identity (the vector unit, §23) gives

$$\|\mathbf{v} - x\mathbf{u}\|^2 = \|\mathbf{r}\|^2 + (\hat{x} - x)^2\|\mathbf{u}\|^2 \;\geq\; \|\mathbf{r}\|^2,$$

with equality only when $x = \hat{x}$. Moving along the line away from the foot of the perpendicular only adds a second, perpendicular leg to the triangle.

Example 46.1. $\mathbf{u} = [\,1, 2\,]$, $\mathbf{v} = [\,3, 1\,]$: $\hat{x} = \frac{3+2}{1+4} = 1$, so the closest point is $[\,1, 2\,]$, the residual is $[\,2, -1\,]$, which is orthogonal to $\mathbf{u}$, and the smallest distance is $\|[\,2, -1\,]\| = \sqrt{5}$.



Example 46.2 (The best constant)**.** Three measurements $\mathbf{v} = [\,1, 2, 2\,]$, and we want the single number c that best represents them. "Best" in the least-squares sense means minimizing $\|\mathbf{v} - c\,\mathbf{u}\|^2$ with $\mathbf{u} = [\,1, 1, 1\,]$, the sum of the squared differences $(1-c)^2+(2-c)^2+(2-c)^2$. Then $\hat{c} = \frac{1+2+2}{3} = \frac{5}{3}$, the average. The mean of a list of numbers is its least-squares constant.

47. The closest point in a column space

47.1. The problem

Now let there be many unknowns. Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{b} \in \mathbb{R}^m$, typically with $m > n$ and $\mathbf{b} \notin \text{col}[\mathbf{A}]$, so that $\mathbf{Ax} = \mathbf{b}$ has no solution (§45.1). For each $\mathbf{x}$ the **residual** $\mathbf{r} = \mathbf{b} - \mathbf{Ax}$ measures how far the equations are from holding.

Definition 47.1 (Least-squares solution)**.** A **least-squares solution** of $\mathbf{Ax} = \mathbf{b}$ is a vector $\hat{\mathbf{x}}$ that minimizes the length of the residual:

$$\hat{\mathbf{x}} \in \arg\min_{\mathbf{x}\in\mathbb{R}^n} \|\mathbf{Ax} - \mathbf{b}\|_2^2.$$

The name is literal. $\|\mathbf{Ax} - \mathbf{b}\|^2$ is the sum of the squares of the m errors, one per equation, and we make that sum least. With $n = 1$ and $\mathbf{A}$ the single column $\mathbf{u}$, this is §46 exactly.

47.2. The geometry

As $\mathbf{x}$ ranges over $\mathbb{R}^n$, the vector $\mathbf{Ax}$ ranges over the column space (§10.1). So the problem is: *find the point of* col[$\mathbf{A}$] *closest to* $\mathbf{b}$. In §46 the subspace was a line, and the closest point was the foot of the perpendicular. The same is true for any subspace.

Theorem 47.1. *If* $\hat{\mathbf{x}}$ *makes the residual orthogonal to every column of* $\mathbf{A}$*, that is,*

$$\mathbf{A}^\mathsf{T}(\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) = \mathbf{0},$$

then $\hat{\mathbf{x}}$ *is a least-squares solution:* $\|\mathbf{Ax} - \mathbf{b}\| \geq \|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|$ *for every* $\mathbf{x}$*.*

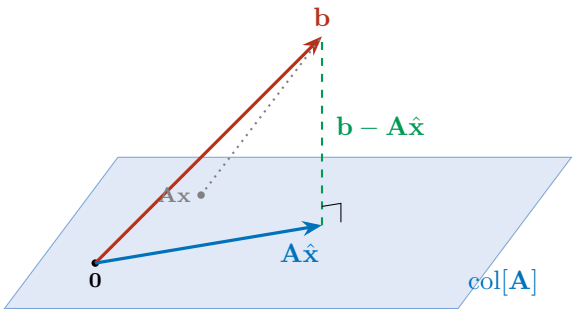
Proof. Write $\mathbf{b} - \mathbf{Ax} = (\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) + \mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})$. The second piece is in col[$\mathbf{A}$], and the first is orthogonal to it:

$$\big(\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\big)^\mathsf{T}(\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) = (\hat{\mathbf{x}} - \mathbf{x})^\mathsf{T}\,\mathbf{A}^\mathsf{T}(\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}) = (\hat{\mathbf{x}} - \mathbf{x})^\mathsf{T}\mathbf{0} = 0.$$

By the Pythagorean identity,

$$\|\mathbf{b} - \mathbf{Ax}\|^2 = \|\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}\|^2 + \|\mathbf{A}(\hat{\mathbf{x}} - \mathbf{x})\|^2 \;\geq\; \|\mathbf{b} - \mathbf{A}\hat{\mathbf{x}}\|^2. \qquad \square$$

Why does the condition read $\mathbf{A}^\mathsf{T}(\cdots) = \mathbf{0}$? A vector is orthogonal to the whole column space exactly when it is orthogonal to each column $\mathbf{a}_j$ (every vector of the column space is a combination of them). The conditions $\mathbf{a}_j^\mathsf{T}\mathbf{r} = 0$, $j = 1, \ldots, n$, are the n entries of $\mathbf{A}^\mathsf{T}\mathbf{r}$, because the rows of $\mathbf{A}^\mathsf{T}$ are the columns of $\mathbf{A}$.



The residual of the best $\hat{\mathbf{x}}$ stands perpendicular to the column space. Any other $\mathbf{Ax}$ in the plane is farther from $\mathbf{b}$, because the dotted segment is the hypotenuse of a right triangle whose leg is the dashed one.

47.3. The normal equations

Multiply out the condition of Theorem 47.1: $\mathbf{A}^\mathsf{T}\mathbf{b} - \mathbf{A}^\mathsf{T}\mathbf{A}\hat{\mathbf{x}} = \mathbf{0}$.

Definition 47.2 (Normal equations)**.** The **normal equations** of the system $\mathbf{A}\mathbf{x} = \mathbf{b}$ are

$$\boxed{\mathbf{A}^\mathsf{T}\mathbf{A}\,\hat{\mathbf{x}} = \mathbf{A}^\mathsf{T}\mathbf{b}.}$$

"Normal" is an old word for perpendicular: these equations say the residual is normal to the column space. They form a *square* system, $n \times n$, even when $\mathbf{A}$ is tall and thin, and its matrix $\mathbf{A}^\mathsf{T}\mathbf{A}$ is the one Part 6 studied. With $n = 1$ and $\mathbf{A} = \mathbf{u}$, they read $(\mathbf{u}^\mathsf{T}\mathbf{u})\hat{x} = \mathbf{u}^\mathsf{T}\mathbf{v}$, which is §46.3.

47.4. The residual lives in the null space of $\mathbf{A}^\mathsf{T}$

The condition $\mathbf{A}^\mathsf{T}\mathbf{r} = \mathbf{0}$ says $\mathbf{r} \in \text{null}[\mathbf{A}^\mathsf{T}]$. By §31.7 applied to $\mathbf{A}^\mathsf{T}$, that null space is orthogonal to the row space of $\mathbf{A}^\mathsf{T}$, which is col[$\mathbf{A}$]. So $\mathbb{R}^m$ splits into the column space and the directions orthogonal to it, and least squares keeps the first piece of $\mathbf{b}$ and discards the second. The map $\mathbf{b} \mapsto \mathbf{A}\hat{\mathbf{x}}$ is an orthogonal projection, onto col[$\mathbf{A}$], and Part 10 writes down its matrix.

48. The same answer by calculus

48.1. What we use

From the vector calculus unit. For a function $f : \mathbb{R}^n \to \mathbb{R}$, the partial derivative $\partial f/\partial x_i$ differentiates in x_i with the other variables held fixed, and the **gradient** ∇f is the vector of all n of them (§6–§7 in two variables; §11–§12 in n). At a minimum of a differentiable function, every partial derivative is zero, so $\nabla f = \mathbf{0}$.

48.2. The function

Expand $f(\mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$ as in §46.2, using M6:

$$f(\mathbf{x}) = (\mathbf{A}\mathbf{x} - \mathbf{b})^\mathsf{T}(\mathbf{A}\mathbf{x} - \mathbf{b}) = \mathbf{x}^\mathsf{T}\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{x} - \mathbf{x}^\mathsf{T}\mathbf{A}^\mathsf{T}\mathbf{b} - \mathbf{b}^\mathsf{T}\mathbf{A}\mathbf{x} + \mathbf{b}^\mathsf{T}\mathbf{b}.$$

The two middle terms are 1×1 matrices, numbers, and each is the transpose of the other, so they are equal:

$$f(\mathbf{x}) = \mathbf{x}^\mathsf{T}\mathbf{M}\mathbf{x} - 2\,\mathbf{c}^\mathsf{T}\mathbf{x} + \mathbf{b}^\mathsf{T}\mathbf{b}, \qquad \mathbf{M} = \mathbf{A}^\mathsf{T}\mathbf{A}, \quad \mathbf{c} = \mathbf{A}^\mathsf{T}\mathbf{b}.$$

A quadratic part, a linear part, and a constant.

48.3. Two gradients

A linear function. $g(\mathbf{x}) = \mathbf{c}^\mathsf{T}\mathbf{x} = c_1x_1 + \cdots + c_nx_n$, so $\partial g/\partial x_i = c_i$ and

$$\nabla(\mathbf{c}^\mathsf{T}\mathbf{x}) = \mathbf{c}.$$

This is the n-variable version of $\frac{d}{dx}(cx) = c$.

A quadratic function. $h(\mathbf{x}) = \mathbf{x}^\mathsf{T}\mathbf{M}\mathbf{x} = \sum_j \sum_k m_{jk}x_jx_k$. Differentiate in x_i. The terms containing x_i are those with $j = i$ or $k = i$:

$$\frac{\partial h}{\partial x_i} = \sum_k m_{ik}x_k + \sum_j m_{ji}x_j = (\mathbf{M}\mathbf{x})_i + (\mathbf{M}^\mathsf{T}\mathbf{x})_i.$$

(The single term with $j = k = i$ is $m_{ii}x_i^2$, whose derivative $2m_{ii}x_i$ is shared between the two sums.) So

$$\nabla(\mathbf{x}^\mathsf{T}\mathbf{M}\mathbf{x}) = (\mathbf{M} + \mathbf{M}^\mathsf{T})\,\mathbf{x} = 2\mathbf{M}\mathbf{x} \quad \text{if } \mathbf{M} \text{ is symmetric,}$$

the n-variable version of $\frac{d}{dx}(mx^2) = 2mx$.

Example 48.1. $\mathbf{M} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$: $h = 2x_1^2 + 2x_1x_2 + 3x_2^2$, so $\nabla h = [\,4x_1 + 2x_2,\ 2x_1 + 6x_2\,]$, which is $2\mathbf{M}\mathbf{x}$. ✓

48.4. Setting the gradient to zero

$\mathbf{M} = \mathbf{A}^\mathsf{T}\mathbf{A}$ is symmetric, so

$$\nabla f(\mathbf{x}) = 2\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{x} - 2\mathbf{A}^\mathsf{T}\mathbf{b}.$$

Setting $\nabla f = \mathbf{0}$ and dividing by 2 gives

$$\mathbf{A}^\mathsf{T}\mathbf{A}\,\hat{\mathbf{x}} = \mathbf{A}^\mathsf{T}\mathbf{b},$$

the normal equations again. Two routes, geometric and analytic, arrive at the same square system.

48.5. A minimum, not a maximum

A zero gradient can also mark a maximum or a saddle. Here it cannot. Along any line $\mathbf{x} + t\mathbf{d}$, f is a one-variable quadratic in t whose t^2 coefficient is

$$\mathbf{d}^\mathsf{T}\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{d} = \|\mathbf{A}\mathbf{d}\|^2 \geq 0,$$

so f curves upward, or is flat, in every direction: its graph is a bowl. Theorem 47.1 had already shown the critical point is a minimum. In the language of the vector calculus unit (§15), the Hessian of f is the constant matrix $2\mathbf{A}^\mathsf{T}\mathbf{A}$, and $\mathbf{d}^\mathsf{T}(2\mathbf{A}^\mathsf{T}\mathbf{A})\mathbf{d} \geq 0$ for every $\mathbf{d}$.

49. Solving the normal equations; fitting a line

49.1. They always have a solution

A system with no solution has been replaced by another system. That only helps if the new one is always solvable, and it is.

Theorem 49.1. *For every $\mathbf{A} \in \mathbb{R}^{m\times n}$ and $\mathbf{b} \in \mathbb{R}^m$, the normal equations $\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{x} = \mathbf{A}^\mathsf{T}\mathbf{b}$ have at least one solution.*

Proof. We need $\mathbf{A}^\mathsf{T}\mathbf{b} \in \text{col}[\mathbf{A}^\mathsf{T}\mathbf{A}]$. Certainly $\mathbf{A}^\mathsf{T}\mathbf{b} \in \text{col}[\mathbf{A}^\mathsf{T}]$. Now $\text{col}[\mathbf{A}^\mathsf{T}\mathbf{A}] \subseteq \text{col}[\mathbf{A}^\mathsf{T}]$, since $\mathbf{A}^\mathsf{T}\mathbf{A}\mathbf{x} = \mathbf{A}^\mathsf{T}(\mathbf{A}\mathbf{x})$, and the two have the same dimension: $\text{rank}(\mathbf{A}^\mathsf{T}\mathbf{A}) = \text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\mathsf{T})$ by §33.2 and §33.3. A subspace inside another of the same dimension is all of it (extend its basis, F2 and F3 of §32.2). So $\text{col}[\mathbf{A}^\mathsf{T}\mathbf{A}] = \text{col}[\mathbf{A}^\mathsf{T}]$, which contains $\mathbf{A}^\mathsf{T}\mathbf{b}$. □

49.2. When the solution is unique

By §31.6, the solutions of the normal equations form one solution plus $\text{null}[\mathbf{A}^\mathsf{T}\mathbf{A}]$, and $\text{null}[\mathbf{A}^\mathsf{T}\mathbf{A}] = \text{null}[\mathbf{A}]$ (§33.1).

Theorem 49.2. *The least-squares solution is unique if and only if $\mathbf{A}$ has full column rank, i.e. its columns are independent.*

If the columns are dependent there are infinitely many least-squares solutions. They all give the *same* closest point $\mathbf{A}\hat{\mathbf{x}}$, since two of them differ by a vector of null[$\mathbf{A}$], which $\mathbf{A}$ sends to $\mathbf{0}$. The point of the column space nearest $\mathbf{b}$ is always unique; only the recipe $\hat{\mathbf{x}}$ for reaching it may not be.

49.3. How to solve them

The normal equations are an ordinary square system, so Part 8 solves them: form $\mathbf{A}^{\mathsf{T}}\mathbf{A}$ and $\mathbf{A}^{\mathsf{T}}\mathbf{b}$, then eliminate and back substitute. With full column rank the elimination finds n pivots and a unique answer.

49.4. Fitting a line

Example 49.1. The data of §45.1: $(1,1)$, $(2,2)$, $(3,2)$. The model $y = c_0 + c_1 t$ gives one equation per point, $\mathbf{A}\mathbf{x} = \mathbf{b}$ with

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} c_0 \\ c_1 \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix},$$

inconsistent by §45.1. Form the normal equations:

$$\mathbf{A}^{\mathsf{T}}\mathbf{A} = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}, \qquad \mathbf{A}^{\mathsf{T}}\mathbf{b} = \begin{bmatrix} 1+2+2 \\ 1+4+6 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix}.$$

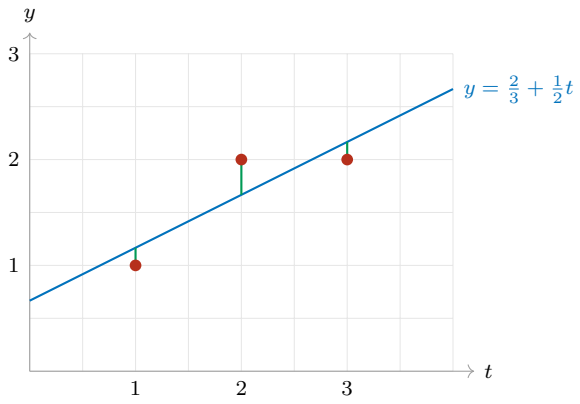
Eliminate: $\left[\begin{array}{cc|c} 3 & 6 & 5 \\ 6 & 14 & 11 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 3 & 6 & 5 \\ 0 & 2 & 1 \end{array}\right]$, so $c_1 = \frac{1}{2}$ and $c_0 = \frac{5 - 6\cdot\frac{1}{2}}{3} = \frac{2}{3}$. The best line is

$$y = \tfrac{2}{3} + \tfrac{1}{2}t.$$

Its predictions at $t = 1, 2, 3$ are $\mathbf{A}\hat{\mathbf{x}} = [\,\frac{7}{6}, \frac{10}{6}, \frac{13}{6}\,]$, and the residual is

$$\mathbf{r} = \mathbf{b} - \mathbf{A}\hat{\mathbf{x}} = \left[\,-\tfrac{1}{6},\ \tfrac{1}{3},\ -\tfrac{1}{6}\,\right], \qquad \mathbf{A}^{\mathsf{T}}\mathbf{r} = \left[\,-\tfrac{1}{6} + \tfrac{1}{3} - \tfrac{1}{6},\ \ -\tfrac{1}{6} + \tfrac{2}{3} - \tfrac{1}{2}\,\right] = [0, 0],$$

orthogonal to both columns, as Theorem 47.1 requires. The sum of the squared errors is $\frac{1}{36} + \frac{4}{36} + \frac{1}{36} = \frac{1}{6}$, and no other line does better.



The green segments are the three entries of the residual. Least squares makes the sum of their squares as small as possible.

49.5. Linear regression

In general, m data points (t_i, y_i) and the model $y = c_0 + c_1 t$ give the $m \times 2$ matrix with rows $[\,1,\ t_i\,]$. Its two columns, all ones and the t_i, are independent unless every t_i is the same, so by Theorem 49.2 the best line is unique. This is **linear regression**, the most used computation in statistics.

The method fits much more than lines. A parabola $y = c_0 + c_1 t + c_2 t^2$ gives rows $[\,1,\ t_i,\ t_i^2\,]$, and any model that is a combination of fixed functions of t with unknown coefficients works the same way. "Linear" refers to the *coefficients* c_j, which enter linearly, not to the shape of the curve.

49.6. Practice

P1. Find the multiple of $\mathbf{u} = [\,2, 1, 2\,]$ closest to $\mathbf{v} = [\,1, 1, 1\,]$, and check that the residual is orthogonal to $\mathbf{u}$.

P2. Fit the line $y = c_0 + c_1 t$ to the points $(0, 6)$, $(1, 0)$, $(2, 0)$ by the normal equations, and check that the residual is orthogonal to both columns.

P3. What is the least-squares constant for the data $2, 4, 9$?

P4. Show that if $\mathbf{b} \in \text{col}[\mathbf{A}]$, every least-squares solution solves $\mathbf{Ax} = \mathbf{b}$ exactly.

P5. Compute $\nabla(\mathbf{x}^\mathsf{T}\mathbf{M}\mathbf{x})$ for the non-symmetric $\mathbf{M} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ directly, and compare with $(\mathbf{M} + \mathbf{M}^\mathsf{T})\mathbf{x}$.

P6. Write $\mathbf{A}$, $\mathbf{A}^\mathsf{T}\mathbf{A}$ and $\mathbf{A}^\mathsf{T}\mathbf{b}$ for fitting the parabola $y = c_0 + c_1 t + c_2 t^2$ to $(-1, 1)$, $(0, 0)$, $(1, 1)$, $(2, 3)$.

Answers

P1. $\hat{x} = \frac{2+1+2}{4+1+4} = \frac{5}{9}$; the closest point is $\frac{5}{9}[\,2, 1, 2\,]$, and the residual is $[\,-\frac{1}{9}, \frac{4}{9}, -\frac{1}{9}\,]$, with $-\frac{2}{9} + \frac{4}{9} - \frac{2}{9} = 0$. ✓

P2. $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}$, $\mathbf{b} = [\,6, 0, 0\,]$; $\mathbf{A}^\mathsf{T}\mathbf{A} = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}$, $\mathbf{A}^\mathsf{T}\mathbf{b} = [\,6, 0\,]$. Elimination gives $2c_1 = -6$, so $c_1 = -3$ and $c_0 = 5$: $y = 5 - 3t$. Residual $[\,6 - 5,\ 0 - 2,\ 0 + 1\,] = [\,1, -2, 1\,]$; its dot products with $[\,1, 1, 1\,]$ and $[\,0, 1, 2\,]$ are 0 and 0. ✓

P3. The mean, 5 (Example 46.2).

P4. Some $\mathbf{x}_0$ has $\mathbf{A}\mathbf{x}_0 = \mathbf{b}$, so the smallest possible value of $\|\mathbf{Ax} - \mathbf{b}\|$ is 0. A least-squares solution attains the smallest value, so $\|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\| = 0$.

P5. $\mathbf{x}^\mathsf{T}\mathbf{M}\mathbf{x} = x_1^2 + 2x_1x_2 + 3x_2^2$, with gradient $[\,2x_1 + 2x_2,\ 2x_1 + 6x_2\,]$; and $\mathbf{M} + \mathbf{M}^\mathsf{T} = \begin{bmatrix} 2 & 2 \\ 2 & 6 \end{bmatrix}$. They agree, and $2\mathbf{Mx}$ would be wrong, since $\mathbf{M}$ is not symmetric.

P6. $\mathbf{A} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}$, $\mathbf{A}^\mathsf{T}\mathbf{A} = \begin{bmatrix} 4 & 2 & 6 \\ 2 & 6 & 8 \\ 6 & 8 & 18 \end{bmatrix}$, $\mathbf{A}^\mathsf{T}\mathbf{b} = [\,5,\ 6,\ 14\,]$ with $\mathbf{b} = [\,1, 0, 1, 3\,]$.

Looking ahead. The normal equations were solved by elimination, because the matrix inverse had not yet been defined in general. Part 10 defines it. Then the least-squares solution has a formula,

$\hat{\mathbf{x}} = (\mathbf{A}^\mathsf{T}\mathbf{A})^{-1}\mathbf{A}^\mathsf{T}\mathbf{b}$, and the matrix in front of $\mathbf{b}$ becomes a generalized inverse in its own right: the pseudo-inverse.